Genomic Features Improve Recurrence Prediction in ER-Positive Breast Cancer: Evidence from METABRIC and TCGA

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Abstract

Background. Recurrence was the least predictable endpoint in our earlier METABRIC study (AUC 0.666 using clinical and treatment features). We asked whether gene-level molecular data improve recurrence prediction, and for which patients.

Methods. We joined the METABRIC clinical table (22 baseline features) with 57 gene-expression gene-set scores, 20 copy-number and mutation driver features, 20 DNA methylation features, and 10 immune and stromal cell estimates. The complete-case cohort comprised 1,295 patients with 545 recurrences (996 ER-positive, 408 recurrences). Penalized Cox regression, random survival forests, logistic regression, LightGBM, FLAML, and MLJAR were evaluated with repeated stratified cross-validation (25–50 splits) in which every data-dependent step was fitted inside the training fold. Gains over the clinical baseline were tested with corrected resampled t-tests and Holm adjustment. Models trained on METABRIC were applied to TCGA-BRCA (902 patients, 100 events) using six harmonized clinical features.

Results. For all patients combined, genomic features did not improve discrimination (Cox C-index 0.649 vs 0.653). In ER-positive patients, gains of 0.02–0.03 were consistent across model families. The tuned Cox model rose from 0.632 to 0.657 (+0.025; improved in 46 of 50 splits; Holm p = 0.14). The tuned random survival forest with a reduced genomic set rose from 0.633 to 0.661 (+0.028, 95% CI 0.009 to 0.046; 45 of 50 splits; Holm p = 0.016). Genomic features did not help predict recurrence after five years, and methylation and immune features added nothing. Fraction of genome altered and proliferation gene sets were the most stable molecular predictors. In TCGA ER-positive patients (679 patients, 66 events), the Cox gain kept its direction (+0.014, 95% CI −0.036 to 0.064), whereas the random survival forest did not transfer (−0.050).

Conclusions. In ER-positive breast cancer, genomic information adds modest but reproducible prognostic information beyond clinical variables, primarily through signals of genomic instability and proliferation. After multiplicity correction, the improvement was statistically significant only for the reduced-feature random survival forest. The size of the gain is like that reported for commercial multigene signatures, and no gain was seen for all patients combined or for late recurrence. Penalized linear models transferred across cohorts and platforms better than tree ensembles. Calibration and clinical utility remain to be established in larger external cohorts with long follow-up.

Keywords: breast cancer; recurrence; METABRIC; TCGA-BRCA; multi-omics; survival analysis; random survival forest; external validation

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1. Introduction

A recurrence is the regrowth of tumor cells that had already disseminated before or around the time of treatment. Clinical features such as tumor size, nodal status, and the Nottingham Prognostic Index (NPI) measure how much disease had the opportunity to spread. They describe the intrinsic behavior of the tumor cells only coarsely, through histologic grade and three receptor markers. Gene-expression assays measure that behavior more directly. Several of them, including the 21-gene Recurrence Score, PAM50 risk of recurrence, EndoPredict, and the Breast Cancer Index, add prognostic information to clinicopathological variables, particularly in ER-positive disease [2–4]. Expression modules associated with outcome also differ by subtype: proliferation dominates in ER+/HER2− tumors and immune response in ER−/HER2− tumors [5].

ER-positive disease is also where clinical need is greatest over the long term. Recurrences continue at a steady rate for at least 20 years after five years of endocrine therapy [6], and a 2025 patient-level meta-analysis showed that extending aromatase-inhibitor therapy beyond five years reduces recurrence by about a quarter at the cost of side effects, so better risk stratification directly informs treatment decisions [7]. METABRIC is the cohort in which the Integrative Clusters were defined [8] and in which copy-number-driven ER-positive subgroups with high late-relapse risk were described [9]. Whole-genome work has since shown that high-risk ER-positive tumors carry complex focal amplifications and structural rearrangements, while typical-risk ER-positive tumors are largely genomically stable [10].

At the same time, benchmark studies caution that multi-omic data often add little to a well-specified clinical model. Across TCGA cancer types, few multi-omic survival methods beat a clinical Cox model, and only slightly [11]; many multi-omic models also degrade when uninformative features are added [12]. Recent large-scale prognostic models in breast cancer that do improve clinical variables have relied on external validation across many cohorts [13]. These findings set two expectations for our design: gains from genomic data should be **modest** (on the order of 0.02 to 0.05 in discrimination), and they should be **concentrated in ER-positive disease**.

In our earlier work, recurrence was the hardest of several METABRIC targets, with a mean AUC of 0.666 from clinical and treatment features [1]. A pilot study (Phase 1) then tested the categorical molecular labels already present in the clinical table (PAM50, IntClust, 3-gene classifier, HER2 copy-number state, cellularity). They added only +0.004 to +0.008 AUC, with slightly larger gains in ER-positive patients and in late recurrence; the labels largely duplicated ER, HER2, and grade information. The present study therefore moves to gene-level data and asks three questions: (1) do expression, copy-number, mutation, methylation, and immune features improve recurrence prediction over the clinical baseline; (2) is any gain concentrated in ER-positive disease or in late recurrence; and (3) does it transfer to an independent cohort?

2. Methods

The study is reported following the TRIPOD+AI guidance for prediction models that use regression or machine learning [14].

2.1 Data sources and cohort

All molecular data were obtained from the cBioPortal *brca_metabric* study [8,15–18] (Table 1). All files share the METABRIC patient identifier (MB-xxxx); each patient has one tumor sample whose identifier equals the

patient's identifier (verified for all 2,509 patients). The final dataset has one row per patient and 141 columns, and a feature dictionary documents every engineered column.

Table 1. Data sources.

Source	Content	Patients
METABRIC clinical table (as in the EGETC paper)	34 clinical, pathology, treatment and outcome columns	2,509
mRNA z-scores (Illumina HT-12)	20,603 genes	1,980
Copy number (GISTIC, −2 to +2)	22,544 genes	1,985
Mutations (173-gene targeted panel)	17,272 mutations	2,433 sequenced
Promoter methylation (RRBS, beta 0–1)	13,188 promoters	1,418

Outcome. Relapse-free status (recurred vs not recurred) and relapse-free months. Unknown outcomes were left missing and never treated as negative.

Modelling cohorts. Patients with a recorded outcome, positive follow-up time, and complete clinical, expression, copy-number, and mutation data: **1,295 patients and 545 recurrences** (996 ER-positive with 408 recurrences). Methylation analyses used 945 of these patients with methylation data. Cohort characteristics are shown in Table 2.

Table 2. Characteristics of the modelling cohort.

Characteristic	All (n = 1,295)	ER-positive (n = 996)	ER-negative (n = 299)
Recurrences, n (%)	545 (42.1)	408 (41.0)	137 (45.8)
Age at diagnosis, median (IQR), years	61.1 (50.8–69.7)	62.9 (53.0–70.6)	54.4 (44.1–65.1)
HER2-positive, %	12.4	7.2	29.4
Lymph-node positive, %	46.9	45.2	52.5
Histologic grade 3, %	52.4	41.7	88.3
Tumor size, median (IQR), mm	22 (17–30)	22 (17–30)	25 (16.5–31.5)
Chemotherapy / hormone therapy / radiotherapy, %	21.9 / 60.5 / 66.9	12.0 / 70.9 / 64.9	54.5 / 25.8 / 73.9
Median follow-up (reverse Kaplan–Meier), months	163	159	174

2.2 Feature engineering

All molecular features were computed **per patient from that patient's own measurements and fixed, published gene lists**. No outcome information and no information from other patients was used, so the features themselves cannot leak test-set information into training.

Clinical baseline (22 features). The same clinical, receptor, and treatment features as the EGETC paper [1]: age, grade, positive nodes, mutation count, NPI, tumor size and stage, ER/PR/HER2 status, chemotherapy, hormone therapy, radiotherapy, mastectomy, menopausal status, and derived indicators (node-positive flag,

high grade, log mutation count, size in cm, NPI group, age group, ER×HER2 interaction). Every comparison is therefore against the published baseline.

Gene-expression gene-set scores (57 features). With about 1,300 patients and 20,000 genes, modelling individual genes invites overfitting and produces results that are hard to interpret. Gene sets group genes that act together in a biological process, compress the data, and make model explanations meaningful. We used all 50 MSigDB Hallmark sets [19,20] (release 2026.1) without hand-picking; each score is the mean z-score of the member genes present in METABRIC (95–100% coverage, median 99%). We also computed the seven Desmedt breast-cancer modules [5], built around prototype genes AURKA (proliferation), ESR1 (ER signaling), ERBB2 (HER2 signaling), STAT1 (immune response), PLAU (invasion), VEGF (angiogenesis), and CASP3 (apoptosis). Each module score is the weighted average of member-gene z-scores using the published coefficients:

$$S_m = \frac{\sum_i w_i z_i}{\sum_i |w_i|}$$

where w_i is the published coefficient and z_i the expression z-score of gene i in module m, with genes matched by Entrez ID (905 of 913 found). Duplicate probes for the same gene were averaged.

Copy-number driver features (11). High-level amplification (GISTIC +2) of nine recurrently amplified breast-cancer drivers, several of which define Integrative Clusters linked to late relapse [8,9]: CCND1 (16.5% of tumors), PAK1 (9.1%), FGFR1 (12.9%), ZNF703 (14.0%), MYC (25.2%), RPS6KB1 (10.3%), PPM1D (9.9%), ERBB2 (15.1%), and MDM2 (3.2%). Deep deletions of CDKN2A, PTEN, RB1, and MAP2K4 are each rare, so they were combined into one indicator (6.8% of tumors). **Fraction of genome altered** (share of genes with any copy-number change) was added as a measure of genomic instability.

Mutation driver features (9). An indicator of any protein-changing (non-silent) mutation in eight frequently mutated drivers [18]: PIK3CA (40.1% of sequenced tumors), TP53 (35.5%), KMT2C (11.5%), GATA3 (11.1%), MAP3K1 (9.7%), CDH1 (9.7%), AKT1 (4.1%), and PTEN (3.9%), plus log mutation burden. Silent, intronic, UTR, and flanking variants were excluded. Sequenced patients without a mutation in a gene were coded 0.

DNA methylation features (20 built, 14 modelled). Motivated by the finding that epigenomic instability in METABRIC tumors correlates with grade, stage, TP53 mutation, and poorer prognosis [21]. Features were global mean promoter methylation, within-tumor spread, fraction of strongly methylated promoters (beta > 0.5), fraction of intermediate values (0.2–0.8) as an epigenomic-disorder proxy, mean promoter methylation of each Desmedt module's genes, and promoter methylation of nine breast-cancer genes. The cBioPortal file contains no normal-tissue samples, so instability was measured within each tumor rather than against normal breast. Single-gene promoters missing in more than 1% of profiled tumors (for example BRCA1 and ESR1) were excluded from modelling to avoid losing patients, leaving 14 methylation features in the models.

Immune and stromal cell estimates (10). Immune composition is prognostic in breast cancer, with regulatory T cells and M0/M2 macrophages linked to poor outcome and CD8+ T cells to better outcome in ER-negative disease [22], and tumor-infiltrating lymphocytes strongly prognostic in triple-negative disease [23]. MCP-counter marker genes [24] were used to score ten populations: T cells, CD8 T cells, cytotoxic lymphocytes, NK cells, B lineage, monocytic lineage, myeloid dendritic cells, neutrophils, endothelial cells, and fibroblasts.

2.3 Redundancy analysis

Many Hallmark sets overlap, and correlated features can split importance between them, inflate model complexity, and mislead interpretation. We measured this redundancy with principal component analysis (PCA) of the 57 standardized gene-set scores across all 1,980 profiled tumors and with hierarchical clustering of their correlation matrix ($|r| \geq$ about 0.7). PCA does not use the outcome, so it describes structure, not predictive value. Whenever principal components were used as model features, PCA was refitted inside each training fold.

2.4 Endpoints and feature sets

Endpoints. (1) Time to recurrence, which uses follow-up time and handles censoring, scored by Harrell's C-index [25]; and (2) binary recurrence, scored by ROC AUC, for comparability with the EGETC paper and because the AutoML tools support only classification. Two subgroups were pre-specified: ER-positive patients (the main question, motivated by the literature above) and late recurrence, analyzed as a landmark among patients still recurrence-free and under follow-up at 60 months, with time measured from month 60.

Feature sets. CLIN (22 features); CLIN+DM (+7 Desmedt modules); CLIN+HM (+50 Hallmark sets); CLIN+PC (+9 principal components of the 57 scores); CLIN+DRIVERS (+11 copy-number and 9 mutation features); CLIN+GEN (+all 77 genomic features, 99 in total); CLIN+PC+DRV (reduced genomic set: 9 principal components + 20 drivers, 51 in total); CLIN+SHAPSEL (+10 genomic features selected by SHAP inside each training fold); CLIN+METH (+14); CLIN+IMM (+10); and combinations. Every comparison within an analysis uses the same patients and the same splits.

2.5 Models

- **Penalized (ridge) Cox regression** (scikit-survival [26]). In the 25-split analyses, the penalty was fixed at $\alpha = 10$; in the tuned analysis it was chosen from {1, 10, 100, 1000} by inner 3-fold cross-validated C-index on the training fold.
- **Random survival forest** (RSF) [27]: 150 trees, $\sqrt{p}$ features per split, minimum leaf size chosen from 15 or 40 by out-of-bag C-index on the training fold.
- **L2 logistic regression** ($C = 0.1$) and **LightGBM** [28] (300 trees, learning rate 0.03, 15 leaves, minimum 30 samples per leaf, row and column subsampling) with fixed settings, for binary recurrence and for SHAP.
- **FLAML** [29] (40 s per fold, internal 3-fold CV; LightGBM, XGBoost, random forest, extra trees, L1 logistic regression) and **MLJAR** [30] (Explain mode as in the EGETC paper, AUC metric, 120 s limit; baseline, linear, decision tree, random forest, XGBoost, and LightGBM candidates). Both tuned only in the training fold.

2.6 Validation and statistical testing

We used stratified 5-fold cross-validation repeated 5 times (25 splits), or 10 times (50 splits) for the tuned survival models, and a single 5-fold split for AutoML because of run time. Everything learned from data (scaling, PCA, penalty and leaf-size tuning, SHAP-based feature selection, AutoML search) happened inside the training fold. Differences from the clinical baseline were tested with the corrected repeated k-fold t-test [31], or the Nadeau–Bengio corrected t-test for single 5-fold CV [32], with Holm adjustment [33] across the comparisons within each analysis. Correlated test folds make the ordinary paired t-test anticonservative, and cross-validation estimates average performance over training sets rather than the error of one fitted model [34]; the corrected tests account for the overlap between training sets. Repeated cross-validation was used to

improve the precision of performance estimates, because the anticipated genomic gain (about 0.02–0.03) was small relative to fold-to-fold variation. A random seed of 42 was used throughout.

2.7 Interpretability

TreeSHAP [35] was computed for LightGBM on CLIN+GEN in each of the 25 test folds; we report mean |SHAP| and how often each feature ranked in the top 15. As an honest test of SHAP-based selection, the top 10 genomic features were also selected by SHAP inside each training fold and evaluated on the held-out fold (CLIN+SHAPSEL).

2.8 External validation in TCGA-BRCA

All results above are internal to METABRIC, a single historical UK/Canadian cohort. TCGA-BRCA [36] (PanCancer Atlas release on cBioPortal; US patients diagnosed mostly in the 2000s–2010s; RNA sequencing instead of microarrays; whole-exome instead of panel sequencing) was used to test whether the genomic gain transfers [37]. The same gene-set, copy-number, and mutation features were built for TCGA with identical definitions.

TCGA does not provide histologic grade, NPI, node counts, chemotherapy or hormone-therapy flags, or receptor immunohistochemistry in this release, so both cohorts were described with six **harmonized clinical features**: age; T category (METABRIC from tumor size: ≤20 mm, ≤50 mm, >50 mm; TCGA from pathologic T, with T4 capped at 3); N category (METABRIC from positive nodes: 0, 1–3, 4–9, ≥10; TCGA from pathologic N); ER status (METABRIC immunohistochemistry; TCGA from a two-component Gaussian mixture on log2 ESR1 RNA-seq expression, which called 75.5% of tumors and 99% of PAM50 luminal tumors ER-positive); ERBB2 high-level amplification; and radiotherapy. Because ERBB2 amplification enters the harmonized clinical set, the genomic block in this analysis has 76 features (HCLIN+GEN, 82 in total). Continuous molecular features were standardized within each cohort to absorb platform differences.

The outcome in TCGA was the progression-free interval (PFI). Using a different endpoint from METABRIC was unavoidable: TCGA does not record relapse-free survival in the same way, and the PanCancer Clinical Data Resource recommends PFI for outcome analyses because it is defined for nearly all patients and accrues more events than overall survival over TCGA's short follow-up, whereas the disease-free interval is defined only for a subset [38]. PFI counts local and distant recurrence, new primary tumors, and progression, so it is the closest available analogue to METABRIC relapse-free survival; any mismatch between the endpoints would be expected to attenuate, not inflate, the external gain. Male patients, patients with metastatic disease at diagnosis, and patients who received neoadjuvant treatment were excluded. Models were tuned and fitted once on all METABRIC patients in each subgroup (relapse-free survival) and applied once to TCGA. Uncertainty was estimated with 1,000 paired bootstrap resamples of the TCGA set.

3. Results

Figure 1 summarizes the main result: adding genomic features improved discrimination in ER-positive patients in every internal analysis, did not help all patients combined or late recurrence, and in TCGA kept its direction for the penalized Cox model but not for the random survival forest. The sections below give the details.

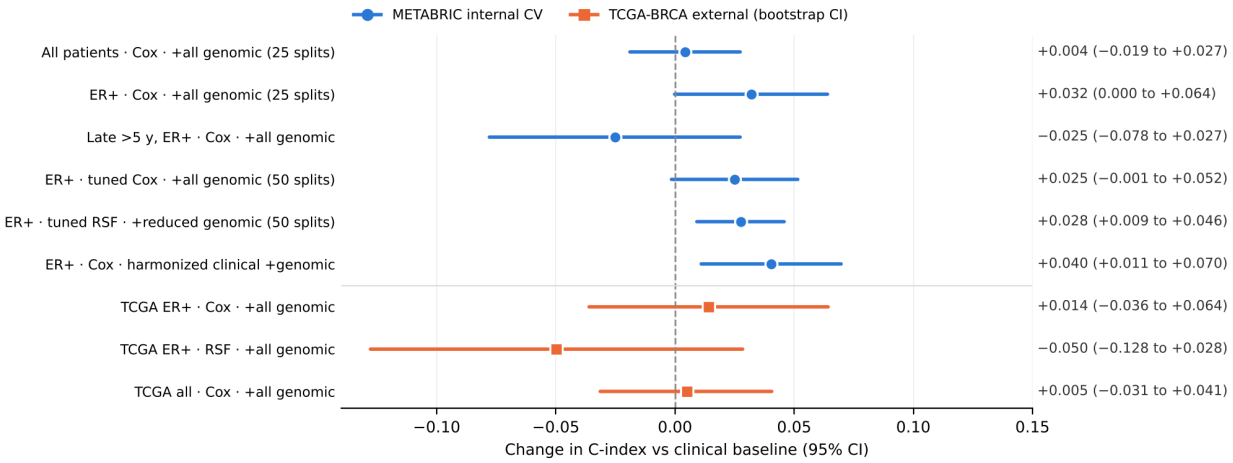


Figure 1. Gain in C-index from adding genomic features to the clinical model. Blue circles: internal METABRIC cross-validation (corrected 95% CI). Orange squares: METABRIC-trained models applied to TCGA-BRCA (bootstrap 95% CI).

3.1 Biological checks of the engineered features

The engineered features behaved as expected. The Desmedt proliferation score rose with grade (−0.46, −0.20, and +0.27 for grades 1–3) and correlated 0.90 with the Hallmark E2F-targets score. The ER-signaling score was higher in ER-positive tumors (+0.21 vs −0.68). ERBB2 amplification was present in 95.1% of HER2-positive tumors and 3.6% of HER2-negative tumors. CCND1 amplification was found in 79% of IntClust 2 tumors, and TP53 mutation was most frequent in basal tumors (88%). PGR promoter methylation was higher in PR-negative tumors, BRCA1 promoter methylation was highest in basal tumors, and CD8 T-cell scores were highest in claudin-low and basal tumors.

3.2 Redundancy among gene-set scores

Across 1,980 tumors, the 57 scores compressed to **9 principal components for 80% of the variance** (18 for 90%). PC1 (24.5%) was an immune and inflammation axis, PC2 (20.3%) a proliferation axis, and PC3 (13.8%) a metabolic and adipose axis. Hierarchical clustering grouped the 57 scores into 32 clusters; the largest were inflammation and immune signaling (8 sets), proliferation (7 sets, including Desmedt AURKA, E2F targets, G2M checkpoint, and MYC targets), invasion and EMT (4 sets, including Desmedt PLAU), and ER signaling (3 sets, including Desmedt ESR1 and both estrogen-response Hallmarks) (Figure 2).

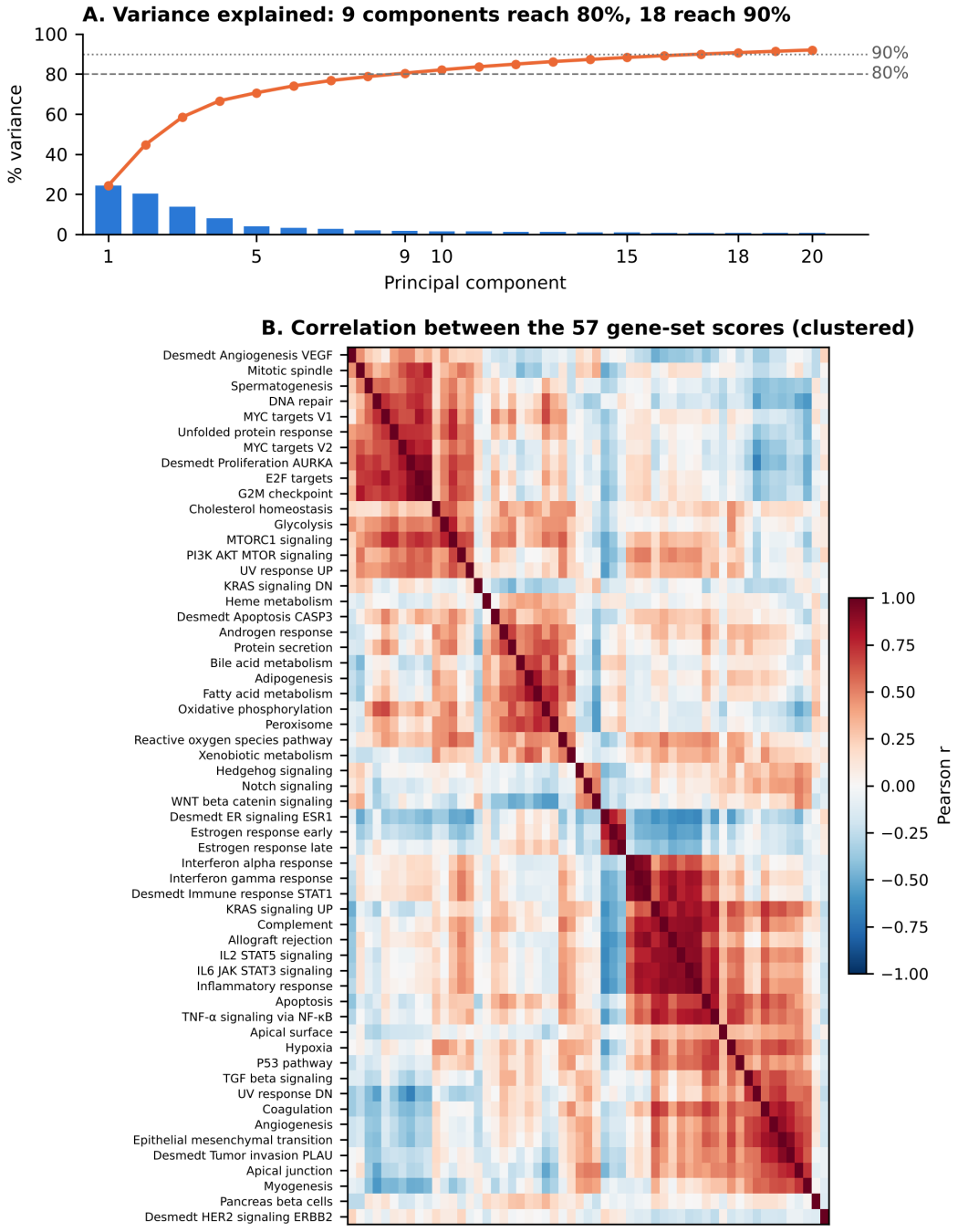


Figure 2. Redundancy among the 57 gene-set scores (1,980 tumors). A: variance explained by principal components (bars) and cumulative variance (line); 9 components reach 80%. B: correlation matrix of the scores, ordered by hierarchical clustering.

3.3 All patients: feature-set comparison

No feature set significantly improved the clinical baseline for all patients combined (Table 3; all Holm-adjusted p = 1.00). The largest gain was +0.011 C-index for CLIN+HM, improved in 18 of 25 splits. SHAP-selected features did not outperform using all features.

Table 3. All patients (n = 1,295; 545 recurrences), 5 × 5 repeated CV. Mean over 25 splits.

Feature set	Features	LR AUC	LightGBM AUC	Cox C-index	Cox gain (splits improved)
CLIN	22	0.660	0.637	0.649	—
CLIN+DM	29	0.654	0.630	0.652	+0.003 (19/25)
CLIN+HM	72	0.663	0.629	0.660	+0.011 (18/25)
CLIN+PC	31	0.657	0.626	0.654	+0.005 (17/25)
CLIN+DRIVERS	42	0.658	0.643	0.652	+0.003 (14/25)
CLIN+GEN	99	0.655	0.636	0.653	+0.004 (15/25)
CLIN+SHAPSEL	32	0.655	0.639	0.653	+0.004 (14/25)

3.4 ER-positive patients and late recurrence

In ER-positive patients, the genomic blocks consistently improved discrimination (Table 4). Hallmark scores added +0.034 C-index (improved in 24 of 25 splits; raw p = 0.018, Holm p = 0.072) and all genomic features +0.032 (23 of 25 splits; Holm p = 0.14). For late recurrence, adding genomic features lowered the C-index: with 188–211 events, models with about 100 features overfit.

Table 4. Penalized Cox (fixed α = 10), C-index, 5 × 5 repeated CV.

Analysis	n (events)	CLIN	+DM	+HM	+DRIVERS	+GEN
All patients	1,295 (545)	0.649	0.652	0.660	0.652	0.653
ER-positive	996 (408)	0.627	0.640	0.661	0.635	0.659
Late (>5 y), all	873 (211)	0.621	0.613	0.601	0.614	0.593
Late (>5 y), ER-positive	703 (188)	0.596	0.587	0.573	0.590	0.571

3.5 AutoML

Both AutoML tools reproduced the pattern (Table 5): no gain for all patients, and gains for most genomic sets in ER-positive patients. The largest were FLAML with drivers (+0.021 AUC) and MLJAR with all genomic features (+0.027), each improved in 4 of 5 folds; FLAML with Hallmark scores was the exception (−0.003). With only five folds, none reached significance. FLAML most often selected LightGBM (21 of 50 fits) or XGBoost (14); MLJAR most often selected its ensemble (37 of 50).

Table 5. AutoML, binary recurrence, ROC AUC, 5-fold CV.

Patients	AutoML	CLIN	+DM	+HM	+DRIVERS	+GEN
All	FLAML	0.662	0.649	0.657	0.666	0.663
All	MLJAR	0.664	0.666	0.660	0.667	0.655
ER-positive	FLAML	0.643	0.648	0.640	0.664	0.655
ER-positive	MLJAR	0.629	0.643	0.640	0.647	0.656

3.6 Features the models relied on

Clinical burden of disease remained the strongest signal (Table 6). Among molecular features, **genomic instability** (fraction of genome altered) was the most stable predictor, ranking in the top 15 in every split, followed by the reactive oxygen species pathway and proliferation-related gene sets (mitotic spindle, G2M checkpoint). When SHAP selected genomic features inside training folds, fraction of genome altered was chosen in 100% of splits, reactive oxygen species in 96%, G2M checkpoint in 84%, and mitotic spindle in 80%.

Table 6. TreeSHAP stability, LightGBM on CLIN+GEN, all patients, 25 test folds.

Feature	Type	Mean |SHAP|	In top 15
Positive lymph nodes	clinical	0.315	100%
Nottingham Prognostic Index	clinical	0.148	96%
Fraction of genome altered	copy number	0.146	100%
Reactive oxygen species pathway	expression	0.127	96%
Mitotic spindle	expression	0.113	84%
Tumour size	clinical	0.109	80%
G2M checkpoint	expression	0.106	92%

3.7 Methylation and immune features

Neither block helped (Table 7). In ER-positive patients, methylation added +0.003 C-index while reducing the usable cohort by about a quarter, and immune scores added −0.001. Immune scores were largely redundant with existing gene-set scores (MCP-counter T cells vs Desmedt immune module r = 0.83; fibroblasts vs Hallmark EMT r = 0.93). The genomic gain in ER-positive patients reappeared in the smaller methylation subset (+0.020, 21 of 25 splits), supporting its robustness.

Table 7. Methylation and immune blocks, penalized Cox (α = 10), C-index, 5 × 5 repeated CV.

Analysis	n (events)	CLIN	+GEN	+METH	+IMM	+GEN+IMM
ER-positive, full cohort	996 (408)	0.627	0.659	—	0.626	0.653
ER-positive, methylation subset	736 (307)	0.625	0.645	0.628	0.626	0.641
All, methylation subset	945 (404)	0.636	0.636	0.638	0.634	0.635

3.8 Tuned survival models

With tuning and 50 splits, the ER-positive gain reached statistical significance for the random survival forest with the reduced genomic set (+0.028, 95% CI +0.009 to +0.046, 45 of 50 splits, Holm p = 0.016; Tables 8 and 9). Feature reduction helped the tree model, where 51 non-redundant features outperformed 99 overlapping ones, but not Cox, whose ridge penalty already handles correlated features. For all patients, gains were +0.009 to +0.012 and not significant. The Cox penalty was often chosen at the top of the grid (α = 1000 in 83% of all-patient fits and 33% of ER-positive fits; α = 100 in 62% of ER-positive fits), suggesting that a wider grid could improve Cox slightly. The RSF chose the smaller leaf size (15) in 66% of the fits.

Table 8. Tuned survival models, C-index, 10 × 5 repeated CV (50 splits).

Patients	Feature set	Features	Tuned Cox	Tuned RSF
ER-positive (n = 996)	CLIN	22	0.632	0.633
ER-positive	CLIN+GEN	99	0.657	0.656
ER-positive	CLIN+PC+DRV	51	0.645	0.661
All (n = 1,295)	CLIN	22	0.653	0.658
All	CLIN+GEN	99	0.663	0.669
All	CLIN+PC+DRV	51	0.662	0.669

Table 9. Gains over CLIN in ER-positive patients (tuned models, 50 splits).

Comparison	Gain	95% CI	Splits improved	Holm p
Cox, CLIN+GEN	+0.025	−0.001 to +0.052	46/50	0.14
Cox, CLIN+PC+DRV	+0.012	−0.008 to +0.032	38/50	0.23
RSF, CLIN+GEN	+0.023	0.000 to +0.046	43/50	0.14
RSF, CLIN+PC+DRV	**+0.028**	**+0.009 to +0.046**	**45/50**	**0.016**

3.9 External validation in TCGA-BRCA

After exclusions, 902 TCGA patients (100 events) had complete data, of whom 679 were ER-positive (66 events); median follow-up was 28 months by reverse Kaplan–Meier (median observed time 25 months). The direction of the ER-positive genomic gain transferred for the ridge Cox model (+0.014), but the gain was about a third of the internal estimate and not statistically distinguishable from zero (Table 10; Figure 1). For reference, the same harmonized comparison inside METABRIC (Cox, 5 × 5 CV) gave a larger ER-positive gain: 0.615 to 0.655 (+0.040, 95% CI +0.011 to +0.070, raw p = 0.009, improved in 23 of 25 splits); the all-patient gain was +0.016 (not significant). The random survival forest did **not** transfer with all 82 features, it performed worse in TCGA than the clinical model.

Table 10. Train on METABRIC, test on TCGA-BRCA. C-index with 1,000 paired bootstrap resamples.

Subgroup (TCGA test set)	Model	Harmonized clinical	+ Genomic	Gain (95% CI)	Resamples with a gain
ER-positive (679; 66 events)	Cox	0.608	0.623	+0.014 (−0.036 to +0.064)	74%
ER-positive	Cox, reduced	0.608	0.615	+0.007 (−0.037 to +0.048)	64%
ER-positive	RSF	0.604	0.554	−0.050 (−0.128 to +0.028)	11%
ER-positive	RSF, reduced	0.604	0.605	0.000 (−0.065 to +0.071)	51%
All (902; 100 events)	Cox	0.649	0.653	+0.005 (−0.031 to +0.041)	59%
All	RSF	0.634	0.613	−0.021 (−0.068 to +0.027)	22%

Reduced = 9 principal components (PCA fitted on METABRIC) + 19 driver features. Harmonized clinical = age, T category, N category, ER status, ERBB2 amplification, radiotherapy.

4. Discussion

4.1 Principal findings

1. **Genomic data add modest, consistent value in ER-positive disease, with partial external support.** Across penalized Cox, random survival forests, FLAML, and MLJAR, adding genomic features to the clinical model improved ER-positive recurrence discrimination by about 0.02–0.03, with improvement in most splits. In TCGA the Cox model kept the direction of the gain (+0.014) with wide uncertainty. The size of the gain matches what published multigene signatures add to clinicopathological factors, and the subgroup matches where those signatures work best [4,5].
2. **No meaningful gain for all patients combined.** In ER-negative tumors, receptor status, grade, and nodes already capture much of the aggressive biology, and the main prognostic molecular signal reported for that group (immune response) [5,22] did not add to our models. This agrees with benchmark studies in which multi-omic models rarely beat a clinical Cox model [11,12].
3. **No gain for late recurrence.** Despite the late-relapse IntClust biology described in METABRIC [9], our landmark analysis did not benefit from genomic features. With about 190–210 late events, models with about 100 features overfit. Rueda et al. also used a multistate model with distant-relapse endpoints and the full cohort, which our relapse-free endpoint does not reproduce. Given the persistent late risk in ER-positive disease [6,7], this remains an important open question that needs larger cohorts or a much smaller, pre-specified feature set.
4. **Genomic instability and proliferation carry molecular signals.** These were the most stable features across splits and selection procedures; their biology is discussed in Section 4.2.
5. **Methylation and immune estimates are largely redundant** with expression gene sets and driver features in this setting.
6. **Linear models transfer better.** The random survival forest gave the strongest internal result but the weakest external one, consistent with tree ensembles learning cohort- and platform-specific structure. The significant internal RSF result should therefore be presented as internal evidence only; the penalized Cox model is the more credible candidate for generalization.
7. **Realistic ceiling.** All discrimination values remain in the 0.63–0.67 range. For comparison, the DeepSurv benchmark on METABRIC reported C-indices of 0.62–0.65 for Cox, DeepSurv, and RSF on overall survival with nine clinical and expression features [39]. Recurrence also depends on factors not measured at diagnosis (treatment adherence, residual disease after surgery), which limits any model built on diagnostic samples. Multimodal models that add digital pathology have recently reached a pooled C-index of about 0.71 across large external cohorts [13], suggesting where further gains may lie.

4.2 Biological interpretation

Fraction of genome altered. The most stable molecular predictor was the share of the genome with copy-number change, a direct readout of chromosomal instability. Chromosomal instability reflects ongoing errors in chromosome segregation and replication, and it generates the copy-number diversity on which selection acts during dissemination and therapy. In METABRIC, the ER-positive Integrative Clusters with the highest late-relapse risk (IntClust 1, 2, 6, and 9) are defined by focal amplifications [8,9]. Whole-genome sequencing has since shown that high-risk ER-positive tumors share a structural archetype with HER2-positive tumors, marked by complex focal and extrachromosomal amplifications linked to ER-driven R-loop formation,

APOBEC3B activity, and replication stress, whereas typical-risk ER-positive tumors are largely genomically stable [10]. Fraction of genome altered is a crude, platform-robust summary of that axis, which may explain why it carried signal in ER-positive disease. Its failure to improve late-recurrence prediction in our landmark analysis suggests that the instability signal is not specific to late relapse, or that a single genome-wide summary is too coarse to capture the amplicon-specific biology described by Rueda et al. [9]. Copy-number and expression models have also recently been shown to predict long-term recurrence-free survival in METABRIC [40].

Proliferation. The G2M checkpoint and mitotic spindle sets, together with E2F targets and the Desmedt AURKA module, form one tightly correlated proliferation cluster (Figure 2). Proliferation is the core of the commercial signatures that outperform clinical variables in ER-positive disease: the 21-gene Recurrence Score includes a proliferation group (Ki-67, AURKA, BIRC5, CCNB1, MYBL2) [2], the PAM50 risk-of-recurrence score includes a dedicated proliferation index [3,41], and proliferation is the dominant prognostic module specifically in ER+/HER2− tumors [5]. Histologic grade captures proliferation only coarsely, and nearly half of our ER-positive patients had grade 2 tumors (Table 2), were continuous proliferation scores in separate tumors with different outcomes. This offers a biological explanation for why the genomic gain was concentrated in ER-positive disease, where grade and receptor status leave the most room for refinement, and absent in ER-negative disease, where nearly 90% of tumors were grade 3 and highly proliferative.

Oxidative stress. The reactive oxygen species pathway was the second most stable expression feature. This Hallmark set mainly contains antioxidant and redox-buffering genes and is correlated with the metabolic cluster rather than with proliferation. Redox adaptation could plausibly support survival of disseminated cells and resistance to therapy, but this finding was not anticipated and should be treated as hypothesis-generating.

4.3 Clinical utility

This study shows a statistical improvement in discrimination, not clinical usefulness. A gain of 0.02–0.03 in C-index changes the ranking of only a minority of patients, and whether it would change decisions, such as extending endocrine therapy beyond five years [7] or offering chemotherapy, depends on how many patients cross a treatment threshold and on whether the predicted risks are well calibrated. We did not perform calibration, reclassification, or decision-curve analysis of net benefit [42], so clinical utility remains untested. A clinically oriented evaluation would compare the clinical and clinical-plus-genomic models at fixed risk thresholds for 5- and 10-year recurrence in ER-positive patients, report the proportion reclassified and the net benefit across plausible thresholds, and benchmark both against an established signature such as the Recurrence Score or PAM50 risk of recurrence in a cohort where those are available.

4.4 Limitations

- **Retrospective cohorts.** METABRIC patients were diagnosed in 1977–2005, before routine trastuzumab and modern endocrine strategies.
- **External validation.** TCGA has short follow-up (median 28 months; 66 ER-positive events), a different but unavoidable endpoint (progression-free interval vs relapse-free survival; Section 2.8), different platforms, and harmonized clinical features that omit grade and NPI; TCGA ER status was inferred from ESR1 expression. Within-cohort standardization of molecular features used the unlabeled TCGA feature distribution, which is common for cross-platform transfer but means the model is not applicable to a single new patient without a reference population.

- **Complete case analysis.** The modelling cohort (1,295 of 2,509) excludes patients missing clinical or genomic data; the excluded patients are not a random subset.
- **Multiple analyses.** Holm adjustment was applied within each analysis, not across all analyses in the project, so the overall type I error across the many feature sets, model families, and subgroups we examined is higher than the per-analysis level. Although multiplicity adjustment was performed within each planned comparison set, we interpret isolated significant findings cautiously because numerous exploratory analyses were conducted. ER-positive recurrence was the pre-specified primary question; other comparisons are exploratory. The reduced feature set (CLIN+PC+DRV) and the tuned models were introduced after the first round of analyses on the same data, so the significant RSF result is not a pre-registered test. A small four-feature set (proliferation, ER, immune, fraction of genome altered) that reached Holm p = 0.047 in ER-positive patients is likewise exploratory, because fraction of genome altered was added after seeing SHAP results on the same data.
- **Discrimination only.** We report discrimination only. Calibration (calibration curves, calibration slope, Brier score) was not assessed, and Harrell's C-index can depend on the censoring distribution [43]; calibration and clinical utility are needed before any clinical use [37].
- **Endpoint.** Relapse-free status combines local and distant relapse and does not model competing risks.
- **Methylation.** The cBioPortal file has no normal-tissue reference, so instability measures are within-tumor approximations of the published definitions.
- **Tuning budgets.** AutoML used a single 5-fold split and short time budgets, and the tuned survival grids were small.
- **Treatment variables** reflect clinical decisions and should not be interpreted causally.

4.5 Future work

1. A deep survival model (for example DeepSurv [39]) with careful tuning, compared with the tuned Cox and random survival forest.
2. A wider penalty grid for Cox and elastic-net Cox as an alternative.
3. Calibration of the ER-positive survival models (calibration curves, calibration slope, and integrated Brier scores at 5 and 10 years), Uno's C-index [43], and decision-curve analysis of net benefit [42].
4. Clinical interpretation of the leading molecular features (genomic instability, proliferation, oxidative stress).
5. Validation in a larger external cohort with longer follow-up; the TCGA validation is directionally supportive but underpowered.
6. Spatial and imaging data, such as METABRIC imaging mass cytometry [44] or digital pathology [13], which capture tumor architecture and immune spatial organization does not present in bulk expression.

5. Conclusion

In ER-positive breast cancer, genomic information adds modest but reproducible prognostic information beyond clinical variables, primarily through signals of genomic instability and proliferation. Gene-level genomic features consistently improved recurrence discrimination over a strong clinical baseline in ER-positive breast cancer, by about 0.02–0.03 in C-index, an increment similar to that of commercial multigene signatures; after multiplicity correction, the improvement was statistically significant only for the reduced-

feature random survival forest. No gain was seen for all patients combined or for late recurrence. The molecular signal is carried mainly by genomic instability and proliferation; methylation and immune estimates add nothing further. The gain kept its direction in an independent cohort for penalized Cox regression but not for random survival forests, so linear models are the more robust choice, and confirmation in larger cohorts with long follow-up is required.

Data and code availability

METABRIC and TCGA-BRCA data are publicly available from cBioPortal (https://www.cbioportal.org/datasets: *brca_metabric* and *brca_tcga_pan_can_atlas_2018*). Gene sets came from MSigDB (Hallmark, release 2026.1), the genefu package (Desmedt modules), and MCPcounter (marker genes). All code, the processed dataset, the feature dictionary, and every result table are available at https://github.com/trinhamcity1/METABRIC_Recurrence_Multiomics; *run_all.sh* runs the full pipeline in order (Python 3.11; numpy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, scikit-survival 0.25.0, LightGBM 4.7.0, SHAP 0.49.1, FLAML 2.7.0, mljar-supervised 1.3.2).

References

1.Nham T, Arteaga-Arteaga HB, Bravo-Ortiz MA, Tabares-Soto R, Hernandez A, Arenas G, Guillen-Rondon P. Multi-target machine learning for breast cancer prognosis and molecular classification using METABRIC. In: *Proceedings of EGETC 2026*. 2026. (Best Paper Award.)

2.Paik S, Shak S, Tang G, et al. A multigene assay to predict recurrence of tamoxifen-treated, node-negative breast cancer. *N Engl J Med* 2004;351:2817–2826.

3.Dowsett M, Sestak I, Lopez-Knowles E, et al. Comparison of PAM50 risk of recurrence score with Oncotype DX and IHC4 for predicting risk of distant recurrence after endocrine therapy. *J Clin Oncol* 2013;31:2783–2790.

4.Sestak I, Buus R, Cuzick J, et al. Comparison of the performance of 6 prognostic signatures for estrogen receptor-positive breast cancer: a secondary analysis of a randomized clinical trial. *JAMA Oncol* 2018;4:545–553.

5.Desmedt C, Haibe-Kains B, Wirapati P, et al. Biological processes associated with breast cancer clinical outcome depend on the molecular subtypes. *Clin Cancer Res* 2008;14:5158–5165.

6.Pan H, Gray R, Braybrooke J, et al. 20-year risks of breast-cancer recurrence after stopping endocrine therapy at 5 years. *N Engl J Med* 2017;377:1836–1846.

7.Early Breast Cancer Trialists' Collaborative Group (EBCTCG). Extending the duration of endocrine treatment for early breast cancer: patient-level meta-analysis of 12 randomized trials of aromatase inhibitors in 22 031 postmenopausal women already treated with at least 5 years of endocrine therapy. *Lancet* 2025;406:603–614.

8.Curtis C, Shah SP, Chin SF, et al. The genomic and transcriptomic architecture of 2,000 breast tumours reveals novel subgroups. *Nature* 2012;486:346–352.

9.Rueda OM, Sammut SJ, Seoane JA, et al. Dynamics of breast-cancer relapse reveal late-recurring ER-positive genomic subgroups. *Nature* 2019;567:399–404.

10.Houlahan KE, Mangiante L, Sotomayor-Vivas C, et al. Complex rearrangements fuel ER+ and HER2+ breast tumors. *Nature* 2025;638:510–518.

11.Herrmann M, Probst P, Hornung R, Jurinovic V, Boulesteix AL. Large-scale benchmark study of survival prediction methods using multi-omics data. *Brief Bioinform* 2021;22:bbaa167.

12.Wissel D, Rowson D, Boeva V. Systematic comparison of multi-omics survival models reveals a widespread lack of noise resistance. *Cell Rep Methods* 2023;3:100461.

13.Witowski J, Zeng KG, Cappadona J, et al. Multi-modal AI for comprehensive breast cancer prognostication. *Nat Commun* 2026;17. doi:10.1038/s41467-026-73088-y.

14.Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 2024;385:e078378.

15.Cerami E, Gao J, Dogrusoz U, et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov* 2012;2:401–404.

16.Gao J, Aksoy BA, Dogrusoz U, et al. Integrative analysis of complex cancer genomics and clinical profiles using the cBioPortal. *Sci Signal* 2013;6:pl1.

17.de Bruijn I, Kundra R, Mastrogiacomo B, et al. Analysis and visualization of longitudinal genomic and clinical data from the AACR Project GENIE Biopharma Collaborative in cBioPortal. *Cancer Res* 2023;83:3861–3867.

18.Pereira B, Chin SF, Rueda OM, et al. The somatic mutation profiles of 2,433 breast cancers refine their genomic and transcriptomic landscapes. *Nat Commun* 2016;7:11479.

19.Liberzon A, Birger C, Thorvaldsdóttir H, Ghandi M, Mesirov JP, Tamayo P. The Molecular Signatures Database hallmark gene set collection. *Cell Syst* 2015;1:417–425.

20.Castanza AS, Recla JM, Eby D, Thorvaldsdóttir H, Bult CJ, Mesirov JP. Extending support for mouse data in the Molecular Signatures Database (MSigDB). *Nat Methods* 2023;20:1619–1620.

21.Batra RN, Lifshitz A, Vidakovic AT, et al. DNA methylation landscapes of 1538 breast cancers reveal a replication-linked clock, epigenomic instability and cis-regulation. *Nat Commun* 2021;12:5406.

22.Ali HR, Chlon L, Pharoah PDP, Markowetz F, Caldas C. Patterns of immune infiltration in breast cancer and their clinical implications: a gene-expression-based retrospective study. *PLoS Med* 2016;13:e1002194.

23.Loi S, Drubay D, Adams S, et al. Tumor-infiltrating lymphocytes and prognosis: a pooled individual patient analysis of early-stage triple-negative breast cancers. *J Clin Oncol* 2019;37:559–569.

24.Becht E, Giraldo NA, Lacroix L, et al. Estimating the population abundance of tissue-infiltrating immune and stromal cell populations using gene expression. *Genome Biol* 2016;17:218.

25.Harrell FE Jr, Lee KL, Mark DB. Multivariable prognostic models: issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. *Stat Med* 1996;15:361–387.

26.Pölsterl S. scikit-survival: a library for time-to-event analysis built on top of scikit-learn. *J Mach Learn Res* 2020;21(212):1–6.

27.Ishwaran H, Kogalur UB, Blackstone EH, Lauer MS. Random survival forests. *Ann Appl Stat* 2008;2:841–860.

28.Ke G, Meng Q, Finley T, et al. LightGBM: a highly efficient gradient boosting decision tree. *Adv Neural Inf Process Syst* 2017;30:3146–3154.

29.Wang C, Wu Q, Weimer M, Zhu E. FLAML: a fast and lightweight AutoML library. *Proc Mach Learn Syst* 2021;3:434–447.

30.Płońska A, Płoński P. MLJAR: state-of-the-art automated machine learning framework for tabular data (mljar-supervised). 2021. https://github.com/mljar/mljar-supervised

31.Bouckaert RR, Frank E. Evaluating the replicability of significance tests for comparing learning algorithms. In: *Advances in Knowledge Discovery and Data Mining (PAKDD 2004)*. LNCS 3056. Springer; 2004:3–12.

32.Nadeau C, Bengio Y. Inference for the generalization error. *Mach Learn* 2003;52:239–281.

33.Holm S. A simple sequentially rejective multiple test procedure. *Scand J Stat* 1979;6:65–70.

34.Bates S, Hastie T, Tibshirani R. Cross-validation: what does it estimate and how well does it do it? *J Am Stat Assoc* 2024;119:1434–1445.

35.Lundberg SM, Erion G, Chen H, et al. From local explanations to global understanding with explainable AI for trees. *Nat Mach Intell* 2020;2:56–67.

36.Cancer Genome Atlas Network. Comprehensive molecular portraits of human breast tumors. *Nature* 2012;490:61–70.

37.Collins GS, Dhiman P, Ma J, et al. Evaluation of clinical prediction models (part 1): from development to external validation. *BMJ* 2024;384:e074819.

38.Liu J, Lichtenberg T, Hoadley KA, et al. An integrated TCGA pan-cancer clinical data resource to drive high-quality survival outcome analytics. *Cell* 2018;173:400–416.e11.

39.Katzman JL, Shaham U, Cloninger A, Bates J, Jiang T, Kluger Y. DeepSurv: personalized treatment recommender system using a Cox proportional hazards deep neural network. *BMC Med Res Methodol* 2018;18:24.

40.Rotbauer M, et al. The role of circulating tumor cell-associated genes in the progression of estrogen receptor-positive breast cancer. *npj Breast Cancer* 2025. doi:10.1038/s41523-025-00874-0.

41.Parker JS, Mullins M, Cheang MCU, et al. Supervised risk predictor of breast cancer based on intrinsic subtypes. *J Clin Oncol* 2009;27:1160–1167.

42.Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Med Decis Making* 2006;26:565–574.

43.Uno H, Cai T, Pencina MJ, D'Agostino RB, Wei LJ. On the C-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data. *Stat Med* 2011;30:1105–1117.

44.Danenberg E, Bardwell H, Zanotelli VRT, et al. Breast tumor microenvironment structures are associated with genomic features and clinical outcome. *Nat Genet* 2022;54:660–669.